

PHYS 143 – Mechanics and Thermal Physics

Motion

*How do we describe where an object is, how it moves,
and how its motion changes?*

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By the end of this week, you should be able to:

1. Describe motion using position, displacement, velocity and acceleration.
2. Interpret motion using diagrams, graphs and equations.
3. Select and apply appropriate kinematic models.
4. Apply kinematics to free fall, projectile, circular and relative motion.

The detailed outcomes are in the checklist at the end of the lecture. **Reading:** Young and Freedman, Ch. 2, §2.1–2.6 and Ch. 3, §3.1–3.5.

Three contacts:

1 DESCRIBE

How do we represent motion?

2 MODEL

How do we describe changing motion mathematically?

3 APPLY

How do the same ideas describe more complex motion?

Practice: concept check → lecturer example
→ independent problem → application problem

Where we start: a five-minute check

Not marked. Answer individually, then we review.

1. Give the signs of the x - and y -components of a 20 m/s vector at 30° above the $-x$ axis.
2. Differentiate $x = 3t^2 - 4t + 2$ and evaluate the result at $t = 2 \text{ s}$.
3. Integrate $a = 6t$, given $v = 0$ at $t = 0$.
4. Solve $4.9t^2 - 12t - 3 = 0$. Which root can be a time?
5. What do the slope and the area under a straight-line graph segment mean?
6. Convert 72 km/h to m/s .

These are the Week 1 and calculus skills this week depends on. A calculus refresher is in the supplementary notes.

1. 150° from $+x$: x negative, y positive $(-17.3, +10.0)$.
2. $dx/dt = 6t - 4$, so 8 m/s at $t = 2 \text{ s}$.
3. $v = 3t^2$.
4. $t = \frac{12 \pm \sqrt{144 + 58.8}}{9.8} = 2.68 \text{ s}$ or -0.23 s . Only the positive root.
5. Slope = rise/run; area = base $\times$ height (triangle: half of it).
6. $72/3.6 = 20 \text{ m/s}$.

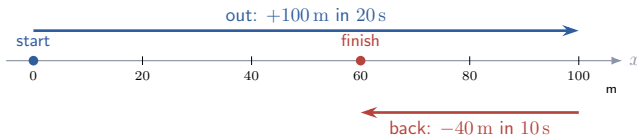
Contact 1: Describe

How do we represent motion?

An opening journey

Opening problem

A runner goes 100 m east in 20 s, turns, and runs 40 m back west in 10 s. Where is the runner? Over the whole journey, how has the position changed, and how rapidly and in what direction?



Predict first

Before any calculation: is the average velocity positive or negative? Is it larger or smaller than the average speed?

Defining the vocabulary of motion

Quantity	What it is	What it measures	SI unit	Representation
Position	location relative to a chosen origin	where the object is	m	a point on an axis; the vector $\vec{r}$
Displacement	final position minus initial position	how far and which way the position has changed	m	arrow from start to finish
Velocity	rate of change of position	how fast, and in which direction, the position changes	m/s	tangent to the path; slope of $x-t$
Speed	the magnitude of the velocity	how fast, with no direction	m/s	length of the velocity arrow
Acceleration	rate of change of velocity	how quickly the speed or the direction changes	m/s ²	arrow; slope of $v-t$

Interpret: velocity answers *how fast, and which way?*; acceleration answers *how is the velocity changing?* Each is defined precisely, with its equation, in the slides that follow.

Position and displacement in one dimension

- **Position** $x(t)$: where the object is, relative to a chosen *origin* along a chosen *axis*.

- **Displacement:**

$$\Delta x = x_2 - x_1 \quad (\text{final} - \text{initial})$$

It can be positive, negative or zero. It does not depend on the path.

- **Distance travelled:** the total path length. Always positive or zero.

Physical meaning

Position depends on the origin and axis you choose.

Displacement is independent of the origin, provided the same coordinate system is used for both positions. Its sign depends on which direction you call positive.

Link to Week 1

In 1D, Δx is the x -component of $\Delta \vec{r} = \vec{r}_2 - \vec{r}_1$, vector subtraction from last week.

Concept check: distance or displacement?

Which of these can be negative? Which is path-independent?

1. The length of road travelled from Accra to Kumasi.
2. The change in position from Accra to Kumasi, with x measured north from Accra.
3. The change in position of a lap runner after one complete lap.
4. The odometer reading minus the starting reading.

Concept check: distance or displacement?

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2. The change in position from Accra to Kumasi, with x measured north from Accra.
3. The change in position of a lap runner after one complete lap.
4. The odometer reading minus the starting reading.

Answers: (1) distance, never negative. (2) displacement, positive here. (3) displacement = 0, distance = one lap. (4) distance. Displacement depends only on end points; distance depends on the whole path.

Definitions

$$v_{\text{av},x} = \frac{\Delta x}{\Delta t} = \frac{x_2 - x_1}{t_2 - t_1}$$

$$\text{average speed} = \frac{\text{distance travelled}}{\Delta t}$$

Velocity is signed. Speed is never negative.

Back to the runner

Find the average velocity and the average speed for the whole 30 s.

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Velocity is signed. Speed is never negative.

Back to the runner

Find the average velocity and the average speed for the whole 30 s.

$$\Delta x = +100 - 40 = +60 \text{ m},$$

$$v_{\text{av},x} = +2.0 \text{ m/s}.$$

Distance = 140 m, so average speed
= 4.7 m/s.

Evaluate: positive, and smaller than the speed. The prediction holds.

Next: what would we need to know to give the velocity at one particular instant?

Question

A dragster is at $x_1 = 19 \text{ m}$ at $t_1 = 1.0 \text{ s}$ and at $x_2 = 277 \text{ m}$ at $t_2 = 4.0 \text{ s}$. Find its average x -velocity.

Predict first

Positive, and roughly what size? (It covers about 250 m in about 3 s.)

Question

A dragster is at $x_1 = 19 \text{ m}$ at $t_1 = 1.0 \text{ s}$ and at $x_2 = 277 \text{ m}$ at $t_2 = 4.0 \text{ s}$. Find its average x -velocity.

Predict first

Positive, and roughly what size? (It covers about 250 m in about 3 s.)

Identify: target $v_{\text{av},x}$; known two positions and times. **Set up:** x along the track, positive forward.

Execute:
$$v_{\text{av},x} = \frac{277 - 19}{4.0 - 1.0} \text{ m/s} = \frac{258}{3.0} = 86 \text{ m/s}.$$

Evaluate: positive, units m/s, consistent with the estimate ($\approx 83 \text{ m/s}$). ✓

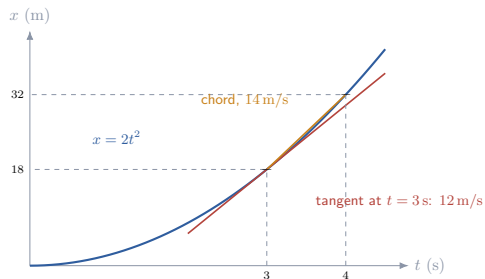
Cf. Young and Freedman, Example 2.1.

From average to instantaneous velocity

A particle moves with $x = 2t^2$ (SI units).
Shrink the interval after $t = 3$ s:

Δt (s)	$v_{\text{av},x}$ (m/s)
1.0	14.0
0.5	13.0
0.1	12.2
0.01	12.02

The chord slope approaches the tangent slope:
12 m/s.



Rule: instantaneous velocity

$$v_x = \lim_{\Delta t \rightarrow 0} \frac{\Delta x}{\Delta t} = \frac{dx}{dt} \quad \text{the slope of the tangent to the } x-t \text{ graph}$$

Reading the graph:

- positive slope $\rightarrow v_x > 0$ (moving in $+x$)
- negative slope $\rightarrow v_x < 0$
- zero slope $\rightarrow$ momentarily at rest
- steeper $\rightarrow$ greater speed $|v_x|$

Two different graphs

An $x-t$ graph is *not* the path of the object. The height of the graph is the position, not the speed.

EQUATION – average velocity

$$v_{\text{av},x} = \frac{\Delta x}{\Delta t} = \frac{x_2 - x_1}{t_2 - t_1}$$

Meaning. the net change of position per unit time over an interval.

Variables. Δx is final minus initial position; Δt is the elapsed time.

Units. m/s.

Interpretation. the sign gives the direction of the net motion; it is not the average speed.

Example. the dragster, +86 m/s; the runner, +2.0 m/s.

EQUATION – instantaneous velocity

$$v_x = \lim_{\Delta t \rightarrow 0} \frac{\Delta x}{\Delta t} = \frac{dx}{dt}$$

Meaning. the velocity at one instant: the limit of the average over ever shorter intervals.

Variables. $x(t)$ is the position as a function of time.

Units. m/s.

Interpretation. the slope of the tangent to the x - t graph; its sign is the direction, its size is the speed.

Example. $x = 2t^2$: $v_x = 4t = 12 \text{ m/s}$ at $t = 3 \text{ s}$, as the table of chords suggested.

Decide, and sketch an $x-t$ graph where it helps

1. Can the average speed over a journey be smaller than $|v_{av,x}|$?
2. Can the velocity be zero while the object is not at the origin?
3. Can the velocity be zero at one instant and the object still be moving a moment later?
4. Can the speed be constant while the velocity changes?

Decide, and sketch an $x-t$ graph where it helps

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Answers

1. **No.** Distance $\geq |\Delta x|$, so average speed $\geq |v_{\text{av},x}|$.
2. **Yes.** Velocity says how position is changing, not where it is.
3. **Yes.** A ball at the top of its flight has $v = 0$ only for an instant.
4. **Yes.** Constant speed on a curve: the direction of $\vec{v}$ changes (Contact 3).

TOOLKIT 1 – Signs and coordinates

CHOOSE: an origin and a positive direction *before* assigning any sign. Mark them on the sketch.

ASSIGN: position, velocity and acceleration are signed; their signs are relative to *your* axis, not to the physics.

INTERPRET: $v_x > 0$ means moving toward $+x$; $v_x < 0$ toward $-x$. Speed is $|v_x|$.

RULE: Change in anything is final minus initial.

WATCH OUT: reversing the axis flips every sign; the physics does not change.

CHECK: does each sign agree with the sketch and the motion described?

From a number line to vectors

In two or three dimensions the position becomes a vector:

$$\vec{r}(t) = x(t) \hat{i} + y(t) \hat{j} + z(t) \hat{k}$$

Rule: displacement and velocity as vectors

$$\Delta \vec{r} = \vec{r}_2 - \vec{r}_1 \quad \vec{v}_{av} = \frac{\Delta \vec{r}}{\Delta t}$$

$$\vec{v} = \frac{d\vec{r}}{dt} = \frac{dx}{dt} \hat{i} + \frac{dy}{dt} \hat{j} + \frac{dz}{dt} \hat{k}$$

Physical meaning

The velocity vector is **tangent to the path** and points in the instantaneous direction of motion. $\vec{v}_{av}$ has the direction of $\Delta \vec{r}$.

Week 1 tools

Components, signs, magnitude

$v = \sqrt{v_x^2 + v_y^2 + v_z^2}$ and
quadrants all carry over.

TOOLKIT 2 – The calculus of motion

position $\vec{r}(t)$ $\xrightarrow{d/dt}$ velocity $\vec{v}(t)$ $\xrightarrow{d/dt}$ acceleration $\vec{a}(t)$

DIFFERENTIATE component by component: $\frac{d}{dt}t^n = nt^{n-1}$, $\frac{d}{dt}\cos\omega t = -\omega\sin\omega t$,

$$\frac{d}{dt}\sin\omega t = \omega\cos\omega t.$$

UNITS: every derivative divides by seconds. Check them at the end.

INTEGRATE (the reverse direction): coming in Contact 2, with initial conditions.

WATCH OUT: the argument of sin and cos is in radians; ωt is dimensionless.

CHECK: does the direction of $\vec{v}$ match the sketch of the path?

Question

A particle has position (SI units, t in seconds)

$$\vec{r} = 2.0 \cos(\pi t) \hat{i} + 2.0 \sin(\pi t) \hat{j} + 1.0 t^2 \hat{k}$$

Find its velocity at $t = 0.5$ s.

Predict first

At $t = 0$ the particle is at $(2.0, 0, 0)$. Sketch the motion. In which direction is it moving at $t = 0.5$ s, and is z increasing?

Identify: target $\vec{v}(0.5\text{ s})$; known $\vec{r}(t)$. **Set up:** differentiate each component (Toolkit 2).

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$$\vec{v} = \frac{d\vec{r}}{dt} = -2.0\pi \sin(\pi t) \hat{i} + 2.0\pi \cos(\pi t) \hat{j} + 2.0t \hat{k}$$

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$$\vec{v} = \frac{d\vec{r}}{dt} = -2.0\pi \sin(\pi t) \hat{i} + 2.0\pi \cos(\pi t) \hat{j} + 2.0t \hat{k}$$

Execute at $t = 0.5\text{ s}$: $\sin(\pi/2) = 1$, $\cos(\pi/2) = 0$, so

$$\vec{v} = (-6.28 \hat{i} + 0 \hat{j} + 1.0 \hat{k}) \text{ m/s}, \quad v = \sqrt{6.28^2 + 1.0^2} = 6.36 \text{ m/s}$$

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Evaluate: at $t = 0.5\text{ s}$ the particle is at $(0, 2.0, 0.25)\text{ m}$, the top of the circle; moving in $-x$ is counter-clockwise from above, and z increases. This agrees with the prediction. Units m/s. ✓

The learning point is differentiating component by component; do not dwell on the geometry. The path is a helix: a circle in the xy -plane combined with steadily increasing z (cf. Young and Freedman, §3.1).

Independent problem

A particle has $\vec{r} = (3.0t^2) \hat{i} + (4.0t) \hat{j}$ (SI units). Find its velocity at $t = 2.0$ s, its speed, and the direction of motion.

Predict first

Sketch first. Is the particle moving mainly in x or y at $t = 2$ s? Speeding up or slowing down in x ?

Independent problem

A particle has $\vec{r} = (3.0t^2) \hat{i} + (4.0t) \hat{j}$ (SI units). Find its velocity at $t = 2.0$ s, its speed, and the direction of motion.

Predict first

Sketch first. Is the particle moving mainly in x or y at $t = 2$ s? Speeding up or slowing down in x ?

$$\vec{v} = 6.0t \hat{i} + 4.0 \hat{j} \Rightarrow \vec{v}(2) = (12 \hat{i} + 4.0 \hat{j}) \text{ m/s. } v = \sqrt{144 + 16} = 12.6 \text{ m/s.}$$
$$\theta = \tan^{-1}(4.0/12) = 18.4^\circ \text{ above } +x \text{ (quadrant I).}$$

Evaluate: mainly x , as predicted; units m/s. ✓

Concept check: reading an $x-t$ graph

Question

Of the four points P, Q, R and S:

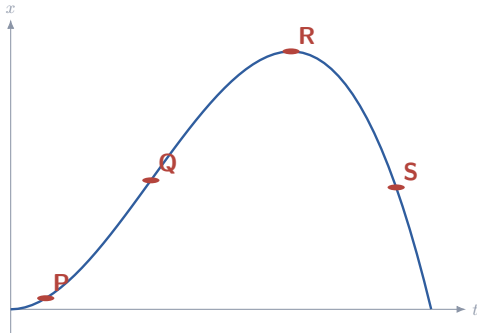
(a) at which is v_x **most positive**?

(b) at which is the **speed** greatest?

Predict first

Decide before the next slide. Which point has the steepest positive slope?

Which the steepest slope of either sign?

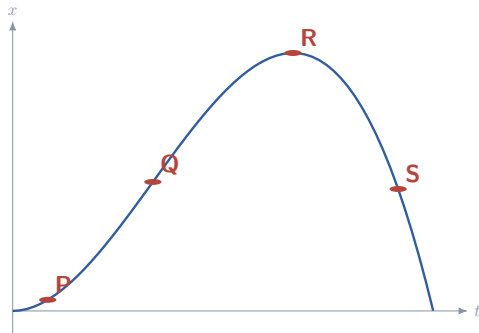


Concept check: solution

This curve is $x = 2t^2 - t^3/3$, so $v_x = 4t - t^2$:

	t (s)	slope	v_x (m/s)
P	0.5	up	+1.75
Q	2	steepest up	+4.0
R	4	flat	0
S	5.5	steepest down	-8.25

(a) Q, (b) S: the speed is $|v_x|$, so a steep *negative* slope counts.



✂ TOOLKIT 3 – Reading motion graphs

Graph	slope gives	area gives
$x-t$	velocity v_x	(not used)
v_x-t	acceleration a_x	displacement Δx (<i>Contact 2</i>)
a_x-t	(rate of change of a)	change in velocity Δv_x (<i>Contact 2</i>)

TRANSLATE both ways: graph $\rightarrow$ description of the motion, and description $\rightarrow$ sketched graph.

RULE: Slope is read from the vertical axis divided by the horizontal axis, with units.

WATCH OUT: the graph is not the path; an upward graph does not mean the object moves upward.

CHECK: does each segment's sign and steepness agree with your motion description?

Common mistakes and the correct habit

Mistake

Reading an $x-t$ graph as the path of the object.

Confusing distance with displacement, speed with velocity.

Assigning signs before choosing the axes.

Using the average velocity as the instantaneous velocity.

Subtracting initial from final the wrong way round.

Instead

The height is the position; the slope is the velocity.

Displacement and velocity are signed; distance and speed are not.

Choose the origin and the positive direction first, and mark them.

Instantaneous velocity is the slope of the tangent, not of the chord.

A change is always final minus initial.

Contact 1: what we can now answer

Where is it? $\rightarrow$ position $\vec{r}(t)$

How has its position changed? $\rightarrow$
displacement $\Delta\vec{r}$

How fast and in what direction is its position changing? $\rightarrow$ velocity $\vec{v} = d\vec{r}/dt$

Next contact

How rapidly is the *velocity* changing? That is acceleration, and with it we can predict motion without a graph.

Connect: Lectures 3 and 4

The slope and area of motion graphs return in Lecture 4, where the area under a force–position graph is the work done. Lecture 3 relates the acceleration you meet next to the force on the object.

Contact 2: Model

How do we describe changing motion mathematically?

Poll

Can an object have a **negative** velocity and still be **speeding up**?

Predict first

A car moves in the $-x$ direction and the driver presses the accelerator. Is its velocity negative? Is it speeding up? Is its acceleration positive or negative?

Poll

Can an object have a **negative** velocity and still be **speeding up**?

Predict first

A car moves in the $-x$ direction and the driver presses the accelerator. Is its velocity negative? Is it speeding up? Is its acceleration positive or negative?

Yes. Negative velocity only says the motion is toward $-x$. Whether the object speeds up depends on how v and a are related; we make this precise in a few slides.

Definition

$$\vec{a}_{\text{av}} = \frac{\Delta \vec{v}}{\Delta t} = \frac{\vec{v}_2 - \vec{v}_1}{t_2 - t_1} \quad \text{unit: m/s}^2$$

In 1D: $a_{\text{av},x} = (v_{2x} - v_{1x})/\Delta t$.

$\Delta \vec{v} = \vec{v}_2 - \vec{v}_1 = \vec{v}_2 + (-\vec{v}_1)$: the same vector subtraction as in Week 1.

Physical meaning

Acceleration describes how the **velocity** changes: its magnitude, its direction, or both. It is not only about speed.

Example

A car's v_x changes from +10 to +30 m/s in 4.0 s:

$$a_{\text{av},x} = +5.0 \text{ m/s}^2.$$

Rule: instantaneous acceleration

$$\vec{a} = \lim_{\Delta t \rightarrow 0} \frac{\Delta \vec{v}}{\Delta t} = \frac{d\vec{v}}{dt} = \frac{d^2 \vec{r}}{dt^2} \quad a_x = \frac{dv_x}{dt} = \frac{d^2 x}{dt^2}$$

- On a v_x - t graph, a_x is the *slope*.
- On an x - t graph, a_x is the *curvature*:
concave up $\rightarrow a_x > 0$; concave down
 $\rightarrow a_x < 0$.

The chain:

$$x(t) \xrightarrow{d/dt} v_x(t) \xrightarrow{d/dt} a_x(t)$$

EQUATION – average acceleration

$$a_{av,x} = \frac{\Delta v_x}{\Delta t} = \frac{v_{2x} - v_{1x}}{t_2 - t_1}$$

Meaning. the change of velocity per unit time over an interval.

Variables. Δv_x is final minus initial velocity; Δt the elapsed time.

Units. m/s^2 , that is, (m/s) per second.

Interpretation. compare its sign with that of v_x : same sign, speeding up; opposite, slowing down.

Example. v_x from $+10$ to $+30 \text{ m/s}$ in 4.0 s : $+5.0 \text{ m/s}^2$.

EQUATION – instantaneous acceleration

$$a_x = \frac{dv_x}{dt} = \frac{d^2x}{dt^2}$$

Meaning. the rate of change of velocity at one instant.

Variables. $v_x(t)$ is the velocity function; $x(t)$ the position function.

Units. m/s^2 .

Interpretation. the slope of the v_x - t graph; the curvature of the x - t graph.

Example. $x = 2t^2$: $v_x = 4t$, $a_x = 4 \text{ m/s}^2$, constant.

Rule: speeding up or slowing down

Rule: when does speed increase?

Same sign of v_x and $a_x \Rightarrow$ speeding up. **Opposite signs** $\Rightarrow$ slowing down.

v_x	a_x	motion
+	+	speeding up (moving toward $+x$)
+	-	slowing down
-	-	speeding up (moving toward $-x$)
-	+	slowing down

Common error

A negative acceleration does not mean slowing down, and a negative velocity does not mean slowing down. Compare the two signs.

Concept check: a motion diagram

Question

The dots show the positions of an object at **equal time intervals**, moving from dot 1 to dot 5 with constant acceleration.

At dot 3, what are the signs of v_x and a_x ?



equal time intervals Δt between dots; the object moves from dot 1 to dot 5

Predict first

Which way is it moving? Are the spacings growing or shrinking?

- The object moves toward $-x$, so $v_x < 0$ at dot 3.
- The spacing between dots **shrinks**, so the speed decreases: slowing down.
- Slowing down means v_x and a_x have **opposite** signs, so $a_x > 0$.

Answer

$v_x < 0$ and $a_x > 0$. Check: v_x goes from a larger negative value to a smaller negative one, so $\Delta v_x > 0$.

Acceleration along and across the path

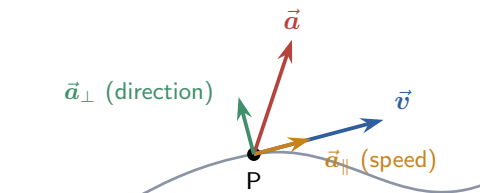
Rule: velocity and the path

$\vec{v}$ is always tangent to the path.

$\vec{a}_{\parallel}$ (parallel to $\vec{v}$) changes the **speed**.

$\vec{a}_{\perp}$ (perpendicular to $\vec{v}$) changes the **direction**.

A skier who speeds up on a straight slope has only $a_{\parallel}$. A car taking a bend at constant speed has only $a_{\perp}$ (Contact 3).



Your turn: translate a $v-t$ graph

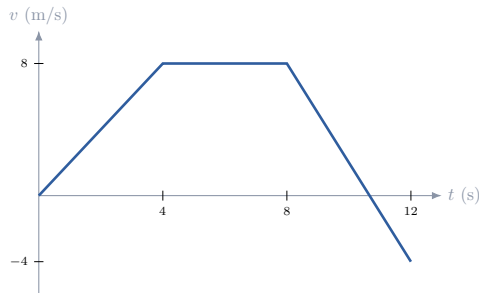
Independent problem

For the motion on the right, find:

- (a) the acceleration in each interval and sketch the a_x-t graph;
- (b) the displacement from 0 to 12 s;
- (c) the distance travelled.

Predict first

Is the displacement larger or smaller than the distance? When does the object reverse?



Solution: slopes and areas

(a) Slopes:

$$0-4\text{ s: } a = +8/4 = +2.0\text{ m/s}^2$$

$$4-8\text{ s: } a = 0$$

$$8-12\text{ s: } a = -12/4 = -3.0\text{ m/s}^2$$

The a_x - t graph is three horizontal steps: +2, 0, -3.

(b), (c) Areas:

$$0-4: \frac{1}{2}(4)(8) = +16\text{ m}$$

$$4-8: 4 \times 8 = +32\text{ m}$$

$$8-10.67: +10.7\text{ m (until } v = 0)$$

$$10.67-12: -2.7\text{ m}$$

Result

Displacement = $16 + 32 + 10.7 - 2.7 = +56\text{ m}$; distance = $16 + 32 + 10.7 + 2.7 = 61.3\text{ m}$.

The object reverses at $t = 10.67\text{ s}$, where v_x crosses zero. Area below the axis is negative displacement but positive distance.

TOOLKIT 3 – Reading motion graphs

Graph	slope gives	area gives
$x-t$	velocity v_x	(not used)
v_x-t	acceleration a_x	displacement Δx
a_x-t	rate of change of a	change in velocity Δv_x

RULE: areas above the axis add; areas below subtract from the displacement but still add to the distance.

WATCH OUT: where $v_x = 0$ the object may reverse, so split the distance calculation there.

CHECK: do the signs and units of the answer match the graph?

TOOLKIT 2 – The calculus of motion (continued)

INTEGRATE to go from acceleration back to position:

$$v_x(t) = v_{0x} + \int_0^t a_x \, dt' \quad x(t) = x_0 + \int_0^t v_x \, dt'$$

The constants matter: v_{0x} and x_0 are the *initial conditions*. Without them the integral is only defined up to a constant.

Area under the graph: integration is the area under a_x - t (giving Δv_x) and v_x - t (giving Δx), Toolkit 3.

WATCH OUT: do not forget the initial value; check $v(0) = v_0$ and $x(0) = x_0$ at the end.

CHECK: differentiate your answer: you must recover the $a(t)$ you started with.

Question

A particle on the x -axis starts at $x_0 = 2.0$ m with $v_{0x} = -3.0$ m/s. Its acceleration is $a_x = (6.0t - 4.0)$ m/s². Find $v_x(t)$, $x(t)$, and the values at $t = 3.0$ s.

Predict first

At $t = 0$, $v_x < 0$ and $a_x < 0$. Is it speeding up or slowing down at first? At $t = 3$ s, what is the sign of a_x ?

Question

A particle on the x -axis starts at $x_0 = 2.0$ m with $v_{0x} = -3.0$ m/s. Its acceleration is $a_x = (6.0t - 4.0)$ m/s². Find $v_x(t)$, $x(t)$, and the values at $t = 3.0$ s.

Predict first

At $t = 0$, $v_x < 0$ and $a_x < 0$. Is it speeding up or slowing down at first? At $t = 3$ s, what is the sign of a_x ?

Execute: $v_x = -3.0 + \int_0^t (6.0t' - 4.0) dt' = -3.0 - 4.0t + 3.0t^2$

$$x = 2.0 + \int_0^t (-3.0 - 4.0t' + 3.0t'^2) dt' = 2.0 - 3.0t - 2.0t^2 + t^3$$

At $t = 3.0$ s: $v_x = -3 - 12 + 27 = +12$ m/s, $x = 2 - 9 - 18 + 27 = 2.0$ m, $a_x = +14$ m/s².

Evaluate: v and a same sign at $t = 0$ (speeding up toward $-x$); v then reverses and the particle returns to $x = 2.0$ m. Differentiating x twice gives $6.0t - 4.0$. ✓

Decide before any equation

1. Can the velocity be zero while the acceleration is not?
2. Can the acceleration be constant while the velocity changes sign?
3. If a_x is constant and non-zero, can v_x stay constant?

Decide before any equation

1. Can the velocity be zero while the acceleration is not?
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Answers

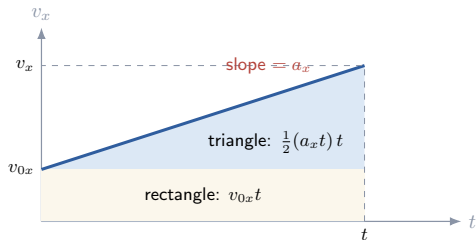
1. **Yes.** At the top of a thrown ball, $v_y = 0$ but $a_y = -g$.
2. **Yes.** Constant $a_x = -2 \text{ m/s}^2$ takes v_x from $+4$ through 0 to -4 m/s .
3. **No.** $a_x = dv_x/dt$; constant velocity means zero acceleration.

When a is constant: skip the calculus

If a_x is constant, $v_x = v_{0x} + a_x t$ is a straight line on a v_x - t graph. The displacement is the area under it:

$$x - x_0 = \underbrace{v_{0x}t}_{\text{rectangle}} + \underbrace{\frac{1}{2}a_x t^2}_{\text{triangle}}$$

Eliminating t between the two results gives the third equation, and the average-velocity form follows from the trapezoid.



Rule: the constant-acceleration equations

Rule: valid only for constant acceleration along a straight line

- | | | |
|-----|--|--------------|
| (1) | $v_x = v_{0x} + a_x t$ | no $x - x_0$ |
| (2) | $x - x_0 = v_{0x} t + \frac{1}{2} a_x t^2$ | no v_x |
| (3) | $v_x^2 = v_{0x}^2 + 2a_x(x - x_0)$ | no t |
| (4) | $x - x_0 = \frac{1}{2}(v_{0x} + v_x) t$ | no a_x |
| (5) | $x - x_0 = v_x t - \frac{1}{2} a_x t^2$ | no v_{0x} |

Five variables appear: $x - x_0$, v_{0x} , v_x , a_x and t . Each equation omits exactly one. Derivations are in the supplementary notes.

EQUATION – equation (1)

$$v_x = v_{0x} + a_x t$$

Meaning. the velocity changes by the same amount every second.

Variables. v_{0x} initial velocity; a_x constant acceleration; t elapsed time.

Units. $\text{m/s} = \text{m/s} + (\text{m/s}^2)(\text{s})$.

Interpretation. a straight line of slope a_x on the v_x - t graph.

Example. the braking example:
 $t = (0 - 25.0)/(-6.0) = 4.2 \text{ s}$.

EQUATION – equation (3)

$$v_x^2 = v_{0x}^2 + 2a_x(x - x_0)$$

Meaning. links the speeds to the displacement, with no time.

Variables. $x - x_0$ is the displacement; the others as in (1).

Units. $\text{m}^2/\text{s}^2 = \text{m}^2/\text{s}^2 + (\text{m/s}^2)(\text{m})$.

Interpretation. the change in v^2 is proportional to the displacement.

Example. the braking example:
 $x - x_0 = 52 \text{ m}$.

Counter-example: the rule has a limit

Apply the constant-acceleration equations to the previous example

There $a_x = 6.0t - 4.0$ varies. At $t = 3.0$ s, $a_x = 14$ m/s². Suppose we wrongly used that value as if it were constant:

Wrong (constant $a = 14$):

$$v_x = -3.0 + 14(3.0) = +39 \text{ m/s}$$

$$x = 2.0 + (-3.0)(3.0) + \frac{1}{2}(14)(9.0) = 56 \text{ m}$$

Correct (integration):

$$v_x = +12 \text{ m/s}$$

$$x = 2.0 \text{ m}$$

Lesson

Before using equations (1)–(5), ask: *is a constant?* If not, integrate.

TOOLKIT 4 – Choosing a constant-acceleration equation

1. Choose the origin and positive direction (Toolkit 1).
2. List the known quantities.
3. Identify the target.
4. Find the variable that is *neither known nor wanted*.
5. Choose the equation that omits it.
6. Solve algebraically first, then substitute with units.
7. Check sign, units and physical meaning.

missing	$x - x_0$	v_x	t	a_x	v_{0x}
use	(1)	(2)	(3)	(4)	(5)

WATCH OUT: matching formulas to symbols without checking that a is constant, or that the motion is on a line.

CHECK: does the answer agree with your prediction and the sketch?

Question

A car moving at 25.0 m/s (90 km/h) brakes with a constant acceleration of magnitude 6.0 m/s^2 until it stops. How far does it travel, and how long does it take?

Predict first

Sketch with $+x$ along the motion. Sign of a_x ? A rough distance if the average speed is about half the initial speed and the stop takes about 4 s ?

Braking distance: solution

Identify: targets $x - x_0$ and t . **Set up:** $+x$ along the motion; $v_{0x} = +25.0 \text{ m/s}$, $v_x = 0$, $a_x = -6.0 \text{ m/s}^2$. Equation (3) omits t ; equation (1) omits $x - x_0$.

Braking distance: solution

Identify: targets $x - x_0$ and t . **Set up:** $+x$ along the motion; $v_{0x} = +25.0 \text{ m/s}$, $v_x = 0$, $a_x = -6.0 \text{ m/s}^2$. Equation (3) omits t ; equation (1) omits $x - x_0$.

Execute: $x - x_0 = \frac{v_x^2 - v_{0x}^2}{2a_x} = \frac{0 - 625}{2(-6.0)} = 52 \text{ m}$ $t = \frac{v_x - v_{0x}}{a_x} = \frac{0 - 25.0}{-6.0} = 4.2 \text{ s}$

Braking distance: solution

Identify: targets $x - x_0$ and t . **Set up:** $+x$ along the motion; $v_{0x} = +25.0 \text{ m/s}$, $v_x = 0$, $a_x = -6.0 \text{ m/s}^2$. Equation (3) omits t ; equation (1) omits $x - x_0$.

Execute: $x - x_0 = \frac{v_x^2 - v_{0x}^2}{2a_x} = \frac{0 - 625}{2(-6.0)} = 52 \text{ m}$ $t = \frac{v_x - v_{0x}}{a_x} = \frac{0 - 25.0}{-6.0} = 4.2 \text{ s}$

Evaluate: average speed $= \frac{1}{2}(25.0 + 0) = 12.5 \text{ m/s}$, and $12.5 \times 4.17 = 52 \text{ m}$, matching equation (4). Units m and s. The estimate ($\approx 50 \text{ m}$) agrees.

Physical meaning

$x - x_0 \propto v_{0x}^2$: doubling the speed quadruples the braking distance.

Rule: free fall

Near the Earth's surface, neglecting air resistance, every object has the same acceleration:

$$a_y = -g = -9.80 \text{ m/s}^2 \quad (+y \text{ upward})$$

It holds **throughout** the motion: rising, at the top, and falling.

Predict first

A hammer and a feather are dropped in vacuum. Which lands first?

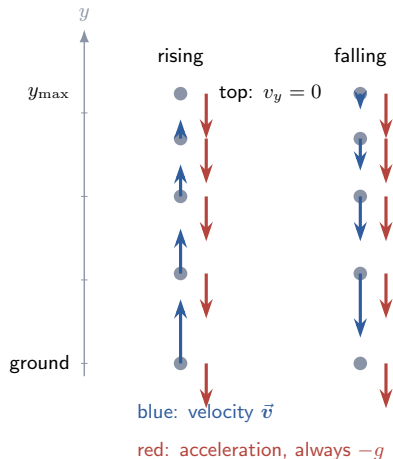
Misconception

“At the top the acceleration is zero.” It is not: $v_y = 0$ momentarily, but $a_y = -g$ still.

Physical meaning

g is the *magnitude*; the sign of a_y comes from the axis you chose (Toolkit 1). With $+y$ downward, $a_y = +g$.

Revisiting the opening throw



A ball is thrown upward at 14.7 m/s from the ground ($+y$ up).

$$\text{Top: } v_y = 0 \Rightarrow t = \frac{14.7}{9.8} = 1.50 \text{ s}$$

$$y_{\max} = \frac{14.7^2}{2(9.8)} = 11.0 \text{ m}$$

Back at ground: $t = 3.00 \text{ s}$ with $v_y = -14.7 \text{ m/s}$.

Neglecting air resistance, and returning to the same height: time up = time down, and the speed is the same at the same height on the way up and down.

Question

A ball is thrown **upward** at 14.7 m/s from the edge of a 30.0 m building. How long until it hits the ground, and what is its velocity then? (Neglect air resistance.)

Predict first

Sketch. Is the time more or less than the 3.00 s of a throw from the ground? Sign of v_y at impact?

Two roots: solution

Identify: targets t and v_y . **Set up:** $+y$ up, origin at the ground: $y_0 = 30.0\text{ m}$, $y = 0$, $v_{0y} = +14.7\text{ m/s}$, $a_y = -9.80\text{ m/s}^2$.

Two roots: solution

Identify: targets t and v_y . **Set up:** $+y$ up, origin at the ground: $y_0 = 30.0\text{ m}$, $y = 0$, $v_{0y} = +14.7\text{ m/s}$, $a_y = -9.80\text{ m/s}^2$.

Execute: (2): $0 = 30.0 + 14.7t - 4.9t^2 \Rightarrow 4.9t^2 - 14.7t - 30.0 = 0$

$$t = \frac{14.7 \pm \sqrt{14.7^2 + 4(4.9)(30.0)}}{2(4.9)} = 4.39\text{ s or } -1.39\text{ s}$$

Two roots: solution

Identify: targets t and v_y . **Set up:** $+y$ up, origin at the ground: $y_0 = 30.0\text{ m}$, $y = 0$, $v_{0y} = +14.7\text{ m/s}$, $a_y = -9.80\text{ m/s}^2$.

Execute: (2): $0 = 30.0 + 14.7t - 4.9t^2 \Rightarrow 4.9t^2 - 14.7t - 30.0 = 0$

$$t = \frac{14.7 \pm \sqrt{14.7^2 + 4(4.9)(30.0)}}{2(4.9)} = 4.39\text{ s or } -1.39\text{ s}$$

Only $t = 4.39\text{ s}$ is physical ($t > 0$). Then $v_y = 14.7 - 9.8(4.39) = -28.4\text{ m/s}$.

Evaluate: longer than 3.00 s as predicted; $v_y < 0$ downward; check (3):

$$v_y^2 = 14.7^2 + 2(9.8)(30.0) = 804, \text{ so } |v_y| = 28.4\text{ m/s } \checkmark$$

Interpretation of the negative root

$t = -1.39\text{ s}$ is the time at which the same parabola would have crossed $y = 0$ *before* the throw. Mathematically valid; physically rejected.

Your turn: a stone from a bridge

Independent problem

A stone is dropped from rest from a bridge and hits the water 2.50 s later. How high is the bridge, and how fast is the stone moving when it hits?

Predict first

Sketch with $+y$ up. Sign of v_y at impact? About how far does it fall in the first second, in the second second?

Your turn: a stone from a bridge

Independent problem

A stone is dropped from rest from a bridge and hits the water 2.50 s later. How high is the bridge, and how fast is the stone moving when it hits?

Predict first

Sketch with $+y$ up. Sign of v_y at impact? About how far does it fall in the first second, in the second second?

$$y - y_0 = v_{0y}t + \frac{1}{2}a_y t^2 = 0 + \frac{1}{2}(-9.80)(2.50)^2 = -30.6 \text{ m, so the bridge is 30.6 m high.}$$
$$v_y = 0 + (-9.80)(2.50) = -24.5 \text{ m/s (downward).}$$

Evaluate: negative displacement and velocity, as predicted; check (3):

$$v_y^2 = 2(9.8)(30.6) = 600, |v_y| = 24.5 \text{ m/s } \checkmark$$

Common mistakes and the correct habit

Mistake

Thinking a negative v or a negative a means slowing down.

Using the constant-acceleration equations when a varies.

Setting $a = 0$ at the top of a flight.

Keeping the quadratic root with $t < 0$.

Integrating without the initial conditions.

Putting $-g$ into an axis that points down.

Instead

Compare the two signs: the same sign means speeding up.

Ask whether a is constant; if not, integrate.

$a_y = -g$ throughout; only v_y is zero there.

Choose the physical root and say why.

Add v_{0x} and x_0 , then check $v(0)$ and $x(0)$.

The sign of a_y comes from the axis you chose.

Contact 2: what we can now do

Position

↓ derivative

Velocity

↓ derivative

Acceleration

and back again by integration with initial conditions.

You should be able to

- decide if speed increases from the signs of v and a ;
- translate graphs and integrate or differentiate;
- choose a constant-acceleration equation and say when it applies;
- solve free fall with a consistent sign convention and the physical root.

Connect: Lectures 3 to 5

$a_x = F_x/m$ in Lecture 3; $v_x^2 = v_{0x}^2 + 2a_x(x - x_0)$ becomes the work–energy theorem in Lecture 4; free fall leads to gravitational potential energy in Lecture 5.

Contact 3: Apply

How do the same ideas describe more complex motion?

An opening puzzle: the monkey and the dart

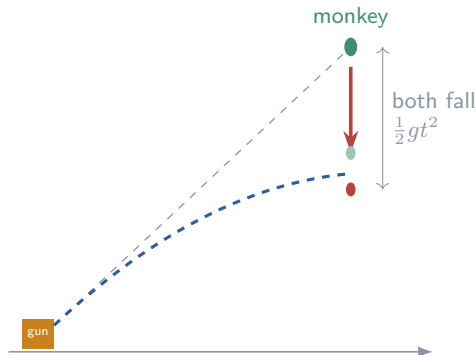
Question

A dart is aimed directly at a monkey hanging in a tree. At the instant the dart leaves the gun, the monkey lets go and falls.

Does the dart hit, pass above, or pass below the monkey? (Assume the dart reaches the monkey's line before the monkey reaches the ground.)

Predict first

Commit to an answer. Which of the two experiences gravity?



Rule: independence of components

For projectile motion under uniform gravity, the horizontal and vertical component equations can be solved independently. They are linked **only through the common time t** .

Why the dart hits:

Without gravity, both the dart and the monkey would stay on the line of sight. With gravity, both fall the *same* distance $\frac{1}{2}gt^2$ below it in time t . The dart still meets the monkey.

Misconception

There is no horizontal force on a projectile. $a_x = 0$, so v_x is constant: the ball does not “carry” a force with it.

Projectile motion: the model

Assumptions: neglect air resistance; g constant; flat Earth over the range.

Horizontal ($a_x = 0$):

$$v_x = v_0 \cos \theta$$

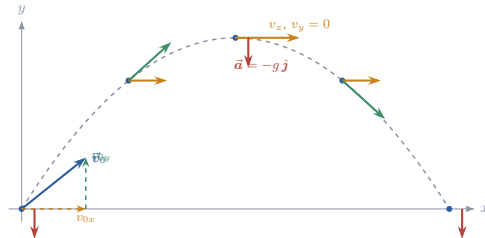
$$x = x_0 + v_0 \cos \theta t$$

Vertical ($a_y = -g$):

$$v_y = v_0 \sin \theta - gt$$

$$y = y_0 + v_0 \sin \theta t - \frac{1}{2}gt^2$$

At the top: $v_y = 0$ but $a_y = -g$.



EQUATION – horizontal motion

$$v_x = v_0 \cos \theta$$

$$x = x_0 + v_0 \cos \theta t$$

Meaning. no horizontal acceleration, so the horizontal velocity is constant.

Variables. v_0 launch speed; θ launch angle above the horizontal; t time.

Units. m/s and m.

Interpretation. the range is this constant speed multiplied by the time of flight.

Example. a long hit:
 $23.0 \text{ m/s} \times 3.94 \text{ s} = 90.4 \text{ m}.$

EQUATION – vertical motion

$$v_y = v_0 \sin \theta - gt$$

$$y = y_0 + v_0 \sin \theta t - \frac{1}{2}gt^2$$

Meaning. constant downward acceleration g acting on the vertical motion alone.

Variables. y_0 launch height; $g = 9.80 \text{ m/s}^2$; $+y$ upward.

Units. m/s and m.

Interpretation. a parabola in t ; $v_y = 0$ only at the top.

Example. a long hit: top at
 $t = 19.3/9.80 = 1.97 \text{ s}.$

Extension: why the projectile trajectory is a parabola

Extension (not assessed). $x(t)$ and $y(t)$ together are the **parametric equations** of the path; t is the parameter. Eliminating t gives the shape of the trajectory.

Launch point at $x_0 = y_0 = 0$

$$y = x \tan \theta - \frac{g}{2v_0^2 \cos^2 \theta} x^2$$

a parabola.

Example

$x = 3.0t$, $y = 4.0t - 4.9t^2$ (SI).
Eliminate t and find where $y = 0$ again.

Extension: why the projectile trajectory is a parabola

Extension (not assessed). $x(t)$ and $y(t)$ together are the **parametric equations** of the path; t is the parameter. Eliminating t gives the shape of the trajectory.

Launch point at $x_0 = y_0 = 0$

$$y = x \tan \theta - \frac{g}{2v_0^2 \cos^2 \theta} x^2$$

a parabola.

Example

$x = 3.0t$, $y = 4.0t - 4.9t^2$ (SI).
Eliminate t and find where $y = 0$ again.

$t = x/3.0$, so $y = 1.33x - 0.544x^2$.
 $y = 0$: $x = 1.33/0.544 = 2.45$ m.

TOOLKIT 5 – Projectile motion

1. **SKETCH** the trajectory and mark the start, top and end. 2. **AXES:** $+x$ horizontal, $+y$ up, origin at launch (Toolkit 1).
 3. **RESOLVE** v_0 into v_{0x} and v_{0y} (Week 1 components, angle from $+x$).
 4. **x -MOTION:** $a_x = 0$. 5. **y -MOTION:** $a_y = -g$ (Toolkit 4 for equations).
 6. **SAME TIME** in both directions; it is the link.
 7. **RECONSTRUCT** the speed and direction if needed (Week 1 Toolkit 4). 8. **CHECK** against the sketch.
- WATCH OUT:** roots of a quadratic in t : choose the physical one; $v_y = 0$ only at the top, not at landing.

Question

A ball is hit from ground level at 30.0 m/s , 40.0° above the horizontal. Find the time to the highest point, the maximum height, the total time of flight and the horizontal range. (Neglect air resistance; the ground is level.)

Predict first

Sketch. At 45° the range would be $v_0^2/g \approx 92 \text{ m}$; is the range at 40° slightly smaller or larger? What are v_y and a_y at the top?

Set up: $v_{0x} = 30.0 \cos 40.0^\circ = 23.0 \text{ m/s}$, $v_{0y} = 30.0 \sin 40.0^\circ = 19.3 \text{ m/s}$.

A long hit: solution

Set up: $v_{0x} = 30.0 \cos 40.0^\circ = 23.0 \text{ m/s}$, $v_{0y} = 30.0 \sin 40.0^\circ = 19.3 \text{ m/s}$.

Execute: top: $v_y = 0 \Rightarrow t = \frac{19.3}{9.80} = 1.97 \text{ s}$; $y_{\max} = \frac{19.3^2}{2(9.80)} = 19.0 \text{ m}$.

Landing: $y = 0 \Rightarrow t = \frac{2(19.3)}{9.80} = 3.94 \text{ s}$ (the other root is $t = 0$).

Range: $x = 23.0 \times 3.94 = 90.4 \text{ m}$.

A long hit: solution

Set up: $v_{0x} = 30.0 \cos 40.0^\circ = 23.0 \text{ m/s}$, $v_{0y} = 30.0 \sin 40.0^\circ = 19.3 \text{ m/s}$.

Execute: top: $v_y = 0 \Rightarrow t = \frac{19.3}{9.80} = 1.97 \text{ s}$; $y_{\max} = \frac{19.3^2}{2(9.80)} = 19.0 \text{ m}$.

Landing: $y = 0 \Rightarrow t = \frac{2(19.3)}{9.80} = 3.94 \text{ s}$ (the other root is $t = 0$).

Range: $x = 23.0 \times 3.94 = 90.4 \text{ m}$.

Evaluate: range slightly less than 92 m, as predicted. At the top $\vec{v} = (23.0 \hat{i} + 0 \hat{j}) \text{ m/s}$ and $a_y = -9.80 \text{ m/s}^2$. Total time is twice the time to the top, as expected for a symmetric path. ✓

Your turn: launched from a height

Independent problem

A ball is kicked at 20.0 m/s , 30.0° above the horizontal, from the edge of a 10.0 m high roof. Find the time to reach the ground, the horizontal distance from the roof, and the velocity (magnitude and direction) at impact.

Predict first

Sketch. Is the time more or less than $2(10.0)/9.80 = 2.04 \text{ s}$? Is the impact velocity above or below the horizontal?

Set up: $v_{0x} = 17.3 \text{ m/s}$, $v_{0y} = 10.0 \text{ m/s}$, $y_0 = 10.0 \text{ m}$, $y = 0$.

Launched from a height: solution

Set up: $v_{0x} = 17.3 \text{ m/s}$, $v_{0y} = 10.0 \text{ m/s}$, $y_0 = 10.0 \text{ m}$, $y = 0$.

Execute: $0 = 10.0 + 10.0t - 4.9t^2 \Rightarrow t = \frac{10.0 \pm \sqrt{100 + 196}}{9.8} = 2.78 \text{ s}$ or -0.74 s .

Range: $17.3 \times 2.78 = 48.1 \text{ m}$.

$v_y = 10.0 - 9.8(2.78) = -17.2 \text{ m/s}$; $v = \sqrt{17.3^2 + 17.2^2} = 24.4 \text{ m/s}$,

$\theta = \tan^{-1} \frac{17.2}{17.3} = 44.8^\circ$ below the horizontal.

Launched from a height: solution

Set up: $v_{0x} = 17.3 \text{ m/s}$, $v_{0y} = 10.0 \text{ m/s}$, $y_0 = 10.0 \text{ m}$, $y = 0$.

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Range: $17.3 \times 2.78 = 48.1 \text{ m}$.

$v_y = 10.0 - 9.8(2.78) = -17.2 \text{ m/s}$; $v = \sqrt{17.3^2 + 17.2^2} = 24.4 \text{ m/s}$,

$\theta = \tan^{-1} \frac{17.2}{17.3} = 44.8^\circ$ below the horizontal.

Evaluate: $2.78 \text{ s} > 2.04 \text{ s}$ as predicted; $v_y < 0$; the landing speed exceeds 20.0 m/s because the ball has fallen 10 m below its launch. ✓

An opening question: acceleration at constant speed?

Poll

A car goes around a circular track at a **constant speed**. Is its acceleration zero?

Predict first

Recall: acceleration is the rate of change of the *velocity vector*. Does the velocity vector stay the same?

An opening question: acceleration at constant speed?

Poll

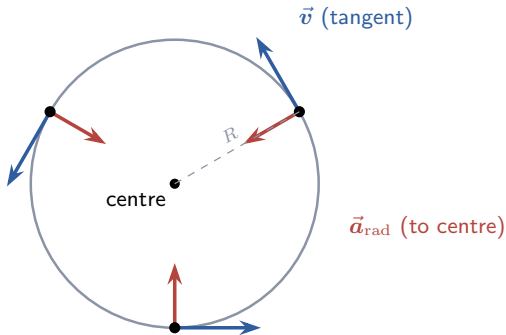
A car goes around a circular track at a **constant speed**. Is its acceleration zero?

Predict first

Recall: acceleration is the rate of change of the *velocity vector*. Does the velocity vector stay the same?

No. The speed is constant but the direction of $\vec{v}$ changes continuously, so $\vec{a} \neq 0$. This is $a_{\perp}$ from Contact 2, with no $a_{\parallel}$.

Uniform circular motion: the geometry

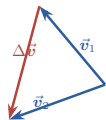
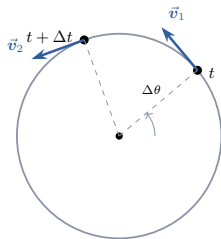


- $\vec{v}$ is tangent to the circle, with constant magnitude v .
- The acceleration has no tangential part: $a_{\text{tan}} = 0$.
- The acceleration points toward the centre: radial, or **centripetal**.

Predict first

If the speed doubles at the same radius, does the acceleration double or quadruple?

Why the acceleration is v^2/R



$\Delta\vec{v}$ points toward the centre

In a short time Δt the velocity turns through an angle $\Delta\theta$. The arc is $v \Delta t = R \Delta\theta$, and $|\Delta\vec{v}| \approx v \Delta\theta$ points toward the centre. So

$$a_{\text{rad}} = \frac{|\Delta\vec{v}|}{\Delta t} = \frac{v \Delta\theta}{\Delta t} = \frac{v \cdot v \Delta t / R}{\Delta t}$$

Rule: radial acceleration

$$a_{\text{rad}} = \frac{v^2}{R}$$

Build the relations from one revolution:

angle 2π rad, distance $2\pi R$

- **Period** T : time for one revolution.
- **Frequency** $f = 1/T$ (hertz, $\text{Hz} = \text{s}^{-1}$).
- **Angular speed** $\omega = 2\pi f = 2\pi/T$ (rad/s).
- **Speed**: $v = \frac{2\pi R}{T} = \omega R$.

Rule: uniform circular motion

$$a_{\text{rad}} = \frac{v^2}{R} = \omega^2 R = \frac{4\pi^2 R}{T^2}$$

Physical meaning

Constant speed does not mean zero acceleration. Also, $a_{\text{rad}} = v^2/R$ gives the radial acceleration at any point of a curved path where the speed is v and the local radius is R .

EQUATION – radial acceleration

$$a_{\text{rad}} = \frac{v^2}{R} = \omega^2 R$$

Meaning. the acceleration that turns the velocity around the circle.

Variables. v speed; R radius; ω angular speed.

Units. m/s^2 ; ω in rad/s .

Interpretation. directed toward the centre; doubling the speed quadruples it.

Example. the bend in the road:
 $20.0^2/80.0 = 5.0 \text{ m/s}^2$.

EQUATION – period, frequency, angular speed

$$T = \frac{1}{f}, \quad \omega = 2\pi f = \frac{2\pi}{T}, \quad v = \omega R$$

Meaning. how often, and how fast in angle, the object goes round.

Variables. T period; f frequency; ω angular speed.

Units. T in s; f in $\text{Hz} = \text{s}^{-1}$; ω in rad/s .

Interpretation. one revolution is 2π rad in time T .

Example. the centrifuge: $f = 50 \text{ Hz}$,
 $\omega = 314 \text{ rad/s}$.

TOOLKIT 6 – Circular motion

VELOCITY is tangent to the path. **ACCELERATION** (uniform motion) points toward the centre.

$$a_{\text{rad}} = \frac{v^2}{R} \quad v = \omega R \quad a_{\text{rad}} = \omega^2 R \quad T = \frac{1}{f} \quad \omega = 2\pi f = \frac{2\pi}{T}$$

SKETCH the circle, mark $\vec{v}$ (tangent) and $\vec{a}$ (to the centre) at one point.

RULE: constant speed does not imply zero acceleration.

WATCH OUT: ω is in rad/s; convert rev/min by $\times 2\pi/60$, and check $v = \omega R$ against $v^2/R = \omega^2 R$.

CHECK: does $\vec{a}$ point to the centre, and is its size reasonable compared with g ?

General curved motion: tangential and normal acceleration

Rule: general curved motion

$$a_t = \frac{dv}{dt} \quad \text{changes the speed}$$

$$a_n = \frac{v^2}{\rho} \quad \text{changes the direction}$$

ρ is the local radius of curvature of the path.
For uniform circular motion $a_t = 0$ and $\rho = R$,
so $a_n = a_{\text{rad}}$.

Link to Contact 2

Tangential acceleration is the component **parallel** to $\vec{v}$ ($a_{\parallel}$);
normal acceleration is the component **perpendicular** to $\vec{v}$ ($a_{\perp}$).

Predict first

A car speeds up while rounding a bend. Which component is non-zero, a_t , a_n or both?

Question

A sample at radius $R = 0.10 \text{ m}$ in a centrifuge rotates at 3000 rev/min . Find the period, the angular speed, the speed of the sample and its radial acceleration in multiples of g .

Predict first

Order of magnitude: much smaller than, similar to, or much larger than g ?

Question

A sample at radius $R = 0.10$ m in a centrifuge rotates at 3000 rev/min. Find the period, the angular speed, the speed of the sample and its radial acceleration in multiples of g .

Predict first

Order of magnitude: much smaller than, similar to, or much larger than g ?

$$f = 3000/60 = 50 \text{ Hz}, \quad T = 1/f = 0.020 \text{ s}, \quad \omega = 2\pi f = 314 \text{ rad/s}, \quad v = \omega R = 31.4 \text{ m/s}.$$
$$a_{\text{rad}} = \omega^2 R = (314)^2(0.10) = 9.9 \times 10^3 \text{ m/s}^2 \approx 1.0 \times 10^3 g.$$

Evaluate: check $v^2/R = 31.4^2/0.10 = 9.9 \times 10^3 \checkmark$; far larger than g , as expected for a centrifuge.

Your turn: a bend in the road

Independent problem

A car travels at a constant 20.0 m/s around a circular bend of radius 80.0 m . Find its radial acceleration, the angular speed, and the time to complete a full circle (if the road were a circle).

Predict first

Draw the direction of $\vec{v}$ and $\vec{a}$ at one point on the circle. Is a larger or smaller than g ?

Independent problem

A car travels at a constant 20.0 m/s around a circular bend of radius 80.0 m. Find its radial acceleration, the angular speed, and the time to complete a full circle (if the road were a circle).

Predict first

Draw the direction of $\vec{v}$ and $\vec{a}$ at one point on the circle. Is a larger or smaller than g ?

$$a_{\text{rad}} = v^2/R = 400/80.0 = 5.0 \text{ m/s}^2 \text{ (toward the centre), about } 0.5 g. \quad \omega = v/R = 0.25 \text{ rad/s.}$$
$$T = 2\pi R/v = 25.1 \text{ s.}$$

Evaluate: check $\omega^2 R = 0.0625 \times 80.0 = 5.0 \text{ m/s}^2 \checkmark$

An opening situation: whose velocity?

Discussion

A passenger walks along the aisle of a train moving at 20.0 m/s relative to the ground. How fast is the passenger moving?

Predict first

The passenger walks toward the front at 1.5 m/s relative to the train. Faster or slower than the train, relative to the ground? What if the passenger walks toward the rear?

An opening situation: whose velocity?

Discussion

A passenger walks along the aisle of a train moving at 20.0 m/s relative to the ground. How fast is the passenger moving?

Predict first

The passenger walks toward the front at 1.5 m/s relative to the train. Faster or slower than the train, relative to the ground? What if the passenger walks toward the rear?

The same motion has different velocities for different observers. We need a notation that says *whose* velocity, and relative to *what*.

Rule: relative velocity

$$\vec{v}_{P/A} = \vec{v}_{P/B} + \vec{v}_{B/A}$$

Read $\vec{v}_{P/A}$ as “the velocity of P relative to A ”.
Also $\vec{v}_{A/B} = -\vec{v}_{B/A}$.

TOOLKIT 7 – Relative velocity

CHAIN: P relative to B , plus B relative to A , equals P relative to A . The inner subscript cancels.

1. Identify the object and the observers.
2. Write the relation *before* inserting values.
3. In 2D, use components (Week 1) and sketch the vector triangle.

WATCH OUT: do not add speeds: add vectors.

CHECK: does the direction agree with the sketch?

EQUATION – relative velocity

$$\vec{v}_{P/A} = \vec{v}_{P/B} + \vec{v}_{B/A}$$

Meaning. the velocity of P seen from A is its velocity seen from B plus the velocity of B seen from A .

Variables. P the moving object; A and B two observers or frames.

Units. m/s for every term.

Interpretation. velocities are vectors, so they add as vectors; the inner subscript B cancels.

Example. the passenger on the train, then the river crossing.

Predict first

The passenger walks at 1.5 m/s toward the rear of a train moving at 20.0 m/s. Predict $v_{P/G}$ before using the rule.

Back to the train

Passenger P , train T , ground G .

Train relative to ground:

$$v_{T/G} = +20.0 \text{ m/s.}$$

Passenger relative to train:

$$v_{P/T} = +1.5 \text{ m/s (toward front).}$$

$$v_{P/G} = v_{P/T} + v_{T/G} = 1.5 + 20.0 = 21.5 \text{ m/s.}$$

Toward the rear: $v_{P/T} = -1.5 \text{ m/s}$, so $v_{P/G} = 18.5 \text{ m/s}$: still moving forward relative to the ground, slower than the train.

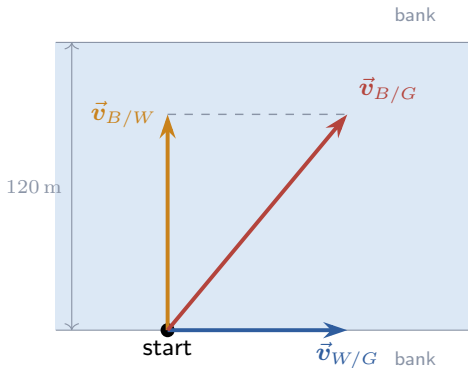
Evaluate: the signs follow from the chosen $+x$ (Toolkit 1).

Question

A river flows east at 3.0 m/s relative to the ground and is 120 m wide. A boat has speed 4.0 m/s relative to the water and is pointed due north. Find its velocity relative to the ground, the crossing time and where it lands.

Predict first

Which way is the boat swept? Is the ground speed more or less than 4.0 m/s ?



Set up: x east, y north. $\vec{v}_{W/G} = 3.0\hat{i}$, $\vec{v}_{B/W} = 4.0\hat{j}$ (m/s). Chain: $\vec{v}_{B/G} = \vec{v}_{B/W} + \vec{v}_{W/G}$.

Set up: x east, y north. $\vec{v}_{W/G} = 3.0 \hat{i}$, $\vec{v}_{B/W} = 4.0 \hat{j}$ (m/s). Chain: $\vec{v}_{B/G} = \vec{v}_{B/W} + \vec{v}_{W/G}$.

Execute: $\vec{v}_{B/G} = (3.0 \hat{i} + 4.0 \hat{j})$ m/s, so $v_{B/G} = 5.0$ m/s at $\tan^{-1}(3.0/4.0) = 36.9^\circ$ east of north.

Crossing time depends only on the northward motion: $t = \frac{120}{4.0} = 30$ s. Downstream drift:
 $3.0 \times 30 = 90$ m.

Crossing a river: solution

Set up: x east, y north. $\vec{v}_{W/G} = 3.0 \hat{i}$, $\vec{v}_{B/W} = 4.0 \hat{j}$ (m/s). Chain: $\vec{v}_{B/G} = \vec{v}_{B/W} + \vec{v}_{W/G}$.

Execute: $\vec{v}_{B/G} = (3.0 \hat{i} + 4.0 \hat{j})$ m/s, so $v_{B/G} = 5.0$ m/s at $\tan^{-1}(3.0/4.0) = 36.9^\circ$ east of north.

Crossing time depends only on the northward motion: $t = \frac{120}{4.0} = 30$ s. Downstream drift:
 $3.0 \times 30 = 90$ m.

Evaluate: swept east, ground speed $5.0 > 4.0$ m/s, as predicted; the current adds sideways motion but does not change the crossing time. $\sqrt{90^2 + 120^2} = 150$ m = 5.0×30 ✓

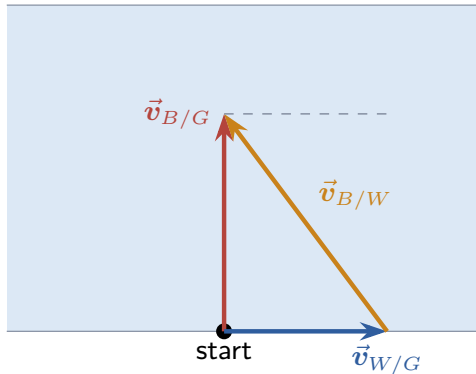
Your turn: heading straight across

Independent problem

The same river (3.0 m/s east, 120 m wide). A boat with speed 5.0 m/s relative to the water must reach the point directly opposite its start. In which direction must it be pointed, and how long does the crossing take?

Predict first

Point the boat upstream or downstream? Is the ground speed more or less than 5.0 m/s?



Heading straight across: solution

Set up: we need $\vec{v}_{B/G}$ due north: $v_{B/G,x} = 0$. From $\vec{v}_{B/G} = \vec{v}_{B/W} + \vec{v}_{W/G}$, with $\vec{v}_{B/W}$ at angle φ west of north:

Heading straight across: solution

Set up: we need $\vec{v}_{B/G}$ due north: $v_{B/G,x} = 0$. From $\vec{v}_{B/G} = \vec{v}_{B/W} + \vec{v}_{W/G}$, with $\vec{v}_{B/W}$ at angle φ west of north:

Execute: x : $-5.0 \sin \varphi + 3.0 = 0 \Rightarrow \sin \varphi = 0.60$, $\varphi = 36.9^\circ$ **west of north** (upstream).

y : $v_{B/G,y} = 5.0 \cos 36.9^\circ = 4.0 \text{ m/s}$; time $t = 120/4.0 = 30 \text{ s}$.

Heading straight across: solution

Set up: we need $\vec{v}_{B/G}$ due north: $v_{B/G,x} = 0$. From $\vec{v}_{B/G} = \vec{v}_{B/W} + \vec{v}_{W/G}$, with $\vec{v}_{B/W}$ at angle φ west of north:

Execute: x : $-5.0 \sin \varphi + 3.0 = 0 \Rightarrow \sin \varphi = 0.60$, $\varphi = 36.9^\circ$ **west of north** (upstream).

y : $v_{B/G,y} = 5.0 \cos 36.9^\circ = 4.0 \text{ m/s}$; time $t = 120/4.0 = 30 \text{ s}$.

Evaluate: upstream, as predicted; the ground speed $4.0 \text{ m/s} < 5.0 \text{ m/s}$ because part of the boat's speed cancels the current. The same triangle as the previous example. ✓

More crosswind problems (cf. Young and Freedman, Examples 3.14 and 3.15) are on the tutorial sheet.

Projectiles

Lecture 5 finds the speed at any height by energy conservation, with no time; Lecture 7 treats collisions in two dimensions with the same component method.

Circular motion

Lecture 3 supplies the force behind $a_{\text{rad}} = v^2/R$ (centripetal force); Lecture 8 extends ω , T and f to rotating bodies.

Relative velocity

Lecture 7: a collision looks different in different frames, and the velocity of the centre of mass is the natural frame in which to describe it.

Physical meaning

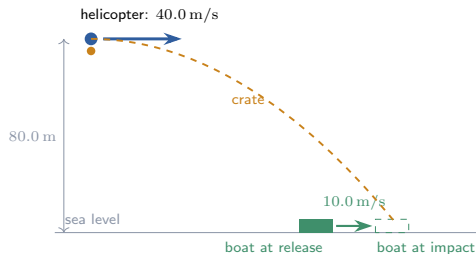
Independent components, the constant-acceleration equations and $\vec{v} = d\vec{r}/dt$ remain the core of the motion toolbox for the rest of the course.

Integrated problem: dropping a crate

Application problem: decide what physics applies

A helicopter flies horizontally at 40.0 m/s , 80.0 m above the sea. A boat below moves in the same direction at 10.0 m/s . A crate released from the helicopter is to land on the boat.

- (a) How far ahead of the crate (horizontally) must the boat be at release?
- (b) With what speed does the crate hit the boat, as seen from the boat?



Predict first

IDENTIFY $\rightarrow$ SET UP (frame? axes? sketch) $\rightarrow$ PREDICT $\rightarrow$ EXECUTE $\rightarrow$ EVALUATE.

Identify: the crate is a projectile (free fall with initial horizontal velocity); the boat is another frame.

Set up: $+x$ forward, $+y$ up; crate $\vec{v}_{C/G} = (40.0, 0)$ at release. Predict: the crate gains horizontally on the boat for the whole fall time.

Identify: the crate is a projectile (free fall with initial horizontal velocity); the boat is another frame.

Set up: $+x$ forward, $+y$ up; crate $\vec{v}_{C/G} = (40.0, 0)$ at release. Predict: the crate gains horizontally on the boat for the whole fall time.

Execute (a): fall time from $y = 80.0$ m: $t = \sqrt{2(80.0)/9.80} = 4.04$ s (Toolkit 5).
Relative horizontal speed $v_{C/B,x} = 40.0 - 10.0 = 30.0$ m/s (Toolkit 7), so the boat must be $30.0 \times 4.04 = 121$ m ahead.

Identify: the crate is a projectile (free fall with initial horizontal velocity); the boat is another frame.

Set up: $+x$ forward, $+y$ up; crate $\vec{v}_{C/G} = (40.0, 0)$ at release. Predict: the crate gains horizontally on the boat for the whole fall time.

Execute (a): fall time from $y = 80.0$ m: $t = \sqrt{2(80.0)/9.80} = 4.04$ s (Toolkit 5).
Relative horizontal speed $v_{C/B,x} = 40.0 - 10.0 = 30.0$ m/s (Toolkit 7), so the boat must be $30.0 \times 4.04 = 121$ m ahead.

(b): $v_{C/B,y} = -9.80(4.04) = -39.6$ m/s, so $v_{C/B} = \sqrt{30.0^2 + 39.6^2} = 49.7$ m/s.

Evaluate: units m and m/s; the crate relative to the boat has the same fall time but a smaller horizontal speed; 4.04 s is plausible for an 80 m fall ($\sqrt{2h/g}$). ✓

Common mistakes and the correct habit

Mistake

A horizontal force on a projectile.

Using different times for x and y .

Setting $a = 0$ at constant speed on a circle.

Adding speeds in relative motion.

Using rev/min or degrees in $v = \omega R$.

Setting $v_y = 0$ at landing.

Instead

There is none: $a_x = 0$ and v_x is constant.

The common time is the only link between the components.

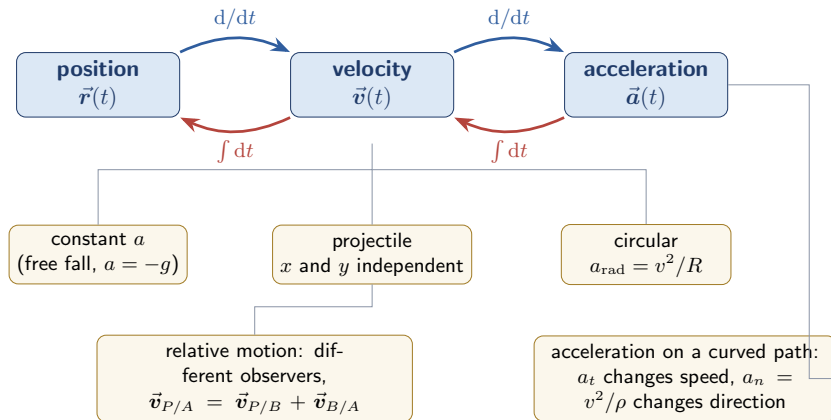
$a = v^2/R$ toward the centre.

Add velocities as vectors with the subscript chain.

Convert to rad/s first.

$v_y = 0$ only at the top of the flight.

Week 2 concept map



MASTER PROCEDURE – MOTION PROBLEMS

1. **IDENTIFY** Which object, which interval, what is asked?
 2. **FRAME** Relative to which observer is the motion described?
 3. **CHOOSE AXES** Origin and positive directions (Toolkit 1).
 4. **SKETCH** Trajectory or motion diagram; add $\vec{v}$ and $\vec{a}$.
 5. **PREDICT** Signs, direction, trend, rough size, physical root.
 6. **SELECT MODEL** General $\vec{r}, \vec{v}, \vec{a}$? Constant a ? Free fall? Projectile? Circular? Relative motion?
 7. **EXECUTE** Apply the matching toolkit; keep signs and units.
 8. **EVALUATE** Units, sign, direction, graph or sketch, assumptions, reasonableness.
- SKETCH $\rightarrow$ PREDICT $\rightarrow$ CALCULATE $\rightarrow$ CHECK.

Inside the course framework

IDENTIFY $\leftrightarrow$ steps 1–2

SET UP $\leftrightarrow$ steps 3–6

EXECUTE $\leftrightarrow$ step 7

EVALUATE $\leftrightarrow$ step 8

Physical meaning

This procedure sits above the seven toolkits. Step 6 tells you which toolkit to open.



THE SEVEN TOOLKITS

- | | | |
|---|--|--|
| 1 | Signs and coordinates | origin and positive direction first; signs relative to your axis; final minus initial |
| 2 | Calculus of motion | differentiate $\vec{r} \rightarrow \vec{v} \rightarrow \vec{a}$; integrate back with initial conditions |
| 3 | Reading motion graphs | slope and area for $x-t$, $v-t$, $a-t$; translate both ways |
| 4 | Choosing a constant-a equation | list known, wanted, missing; confirm a constant |
| 5 | Projectile motion | resolve; independent x and y ; same time; physical root |
| 6 | Circular motion | $\vec{v}$ tangent, $\vec{a}$ to the centre; $a_r = v^2/R = \omega^2 R$ |
| 7 | Relative velocity | subscript chain; vectors not speeds |

The Master Procedure sits above these. Week 1 toolkits (components, reconstructing a vector, adding by components) are reused, not renamed.

Key ideas

- Slope and area link x , v and a graphs.
- Same signs of v and a : speeding up.
- The constant-acceleration equations hold only for constant a .
- $a_y = -g$ throughout the flight, including the top.
- In projectile motion the components are linked only through time.
- Constant speed on a circle still means $a = v^2/R \neq 0$.
- Velocities add as vectors.

Where to look

Definitions: the vocabulary and equation-anatomy slides of each contact.

Common mistakes: the last slide of each contact.

Connections: the Connect boxes and slides.

Summary: core equations

Idea	Equation	Conditions and units
Velocity	$\vec{v} = d\vec{r}/dt, \quad v_{av,x} = \Delta x/\Delta t$	m/s; slope of x - t
Acceleration	$\vec{a} = d\vec{v}/dt = d^2\vec{r}/dt^2$	m/s ² ; slope of v - t
Integration	$v_x = v_{0x} + \int a_x dt, \quad x = x_0 + \int v_x dt$	needs initial conditions
Constant a	$v_x = v_{0x} + a_x t, \quad v_x^2 = v_{0x}^2 + 2a_x(x - x_0)$	only if a is constant, straight line
Free fall	$a_y = -g, \quad g = 9.80 \text{ m/s}^2$	+ y upward; no air resistance
Projectile	$a_x = 0, \quad a_y = -g; \text{ same } t$	$v_x = v_0 \cos \theta, \quad v_{0y} = v_0 \sin \theta$
Circular motion	$a_{rad} = v^2/R = \omega^2 R, \quad v = \omega R$	toward the centre; $\omega = 2\pi f = 2\pi/T$
Curved path	$a_t = dv/dt, \quad a_n = v^2/\rho$	speed and direction changes
Relative velocity	$\vec{v}_{P/A} = \vec{v}_{P/B} + \vec{v}_{B/A}$	vector sum; m/s

Checklist: can I ...

- ☐ distinguish position, displacement, distance, velocity, speed and acceleration?
- ☐ interpret $x-t$, $v-t$ and $a-t$ graphs?
- ☐ relate $\vec{r}$, $\vec{v}$ and $\vec{a}$ using calculus, with initial conditions?
- ☐ interpret the signs of velocity and acceleration?
- ☐ select the correct constant-acceleration model, and say when it fails?
- ☐ solve free-fall problems and choose the physical root?
- ☐ solve projectile problems using independent components?
- ☐ explain and calculate circular acceleration?
- ☐ solve relative-velocity problems?
- ☐ select the appropriate model in an unfamiliar motion problem?

After this week

Tutorial sheet: the train and police-car problems and extra graph, free-fall, projectile, circular and relative-velocity problems. Supplementary notes: derivations, the calculus refresher and the parabola extension.